LNCS Asset Smoke Test

Anonymous

Artifact layout check only

1 Figures

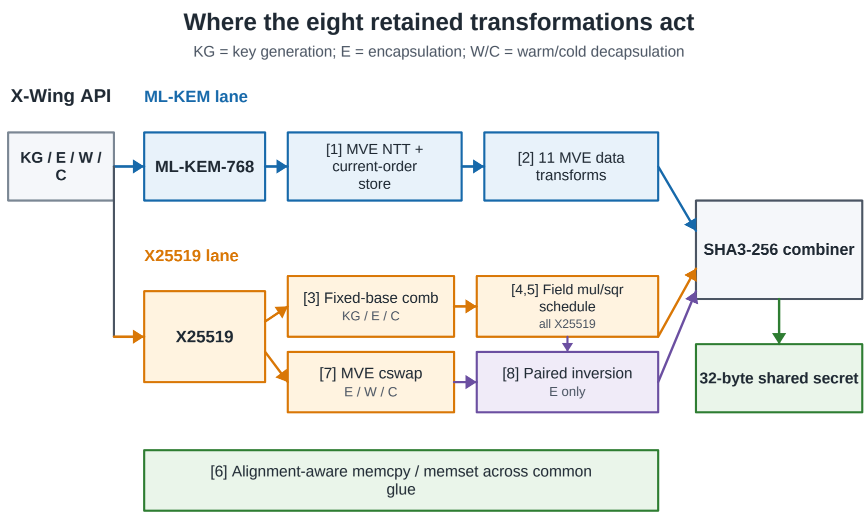

Fig. 1. Placement of the eight retained transformations in the X-Wing data flow. ML-KEM transformations exploit coefficient-level MVE parallelism, whereas X25519 transformations target fixed-base work, field arithmetic, conditional swaps, and the two simultaneous encapsulation endpoints. The alignment-aware memory path is shared glue.

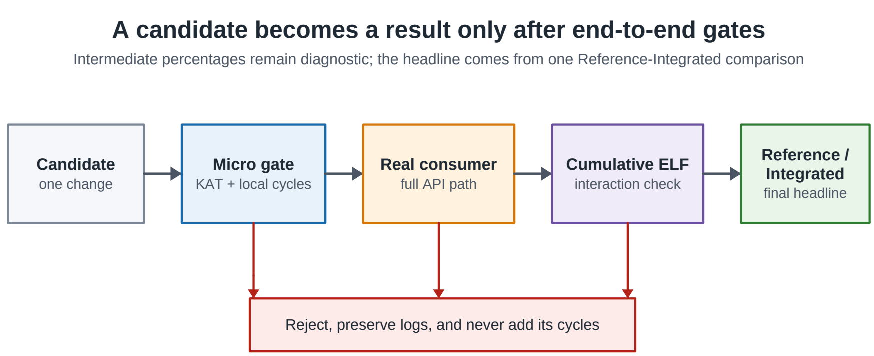

Fig. 2. Adoption pipeline. A candidate must pass correctness and local-cycle gates, improve its real consumer and complete API path, and compose in a cumulative same-ELF build. Intermediate percentages remain diagnostic; the paper's aggregate headline is measured separately by the direct Reference–Integrated comparison.

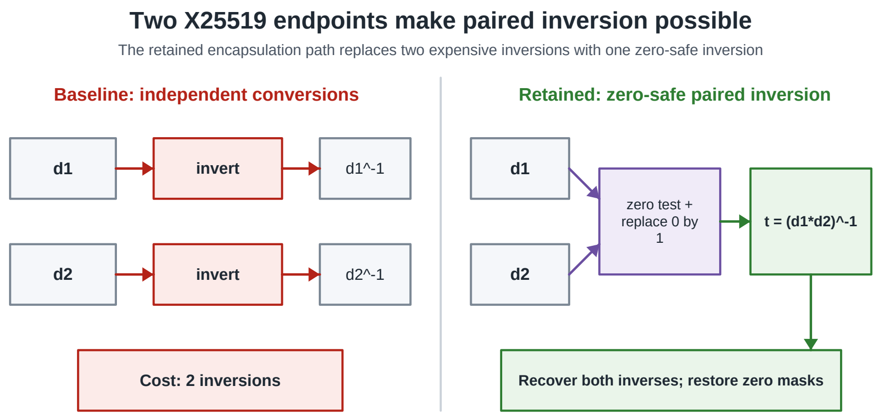

Fig. 3. Independent and paired affine conversion in X-Wing encapsulation. For nonzero denominators, one inversion of $d_1 d_2$ recovers both inverses. Canonical zero tests, replacement by one, and final masks preserve the existing low-order and all-zero behavior.

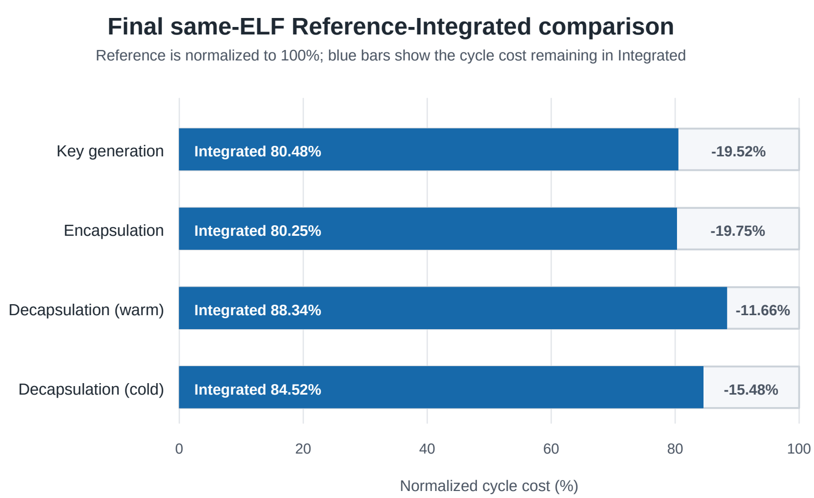

Fig. 4. Direct same-ELF Reference–Integrated result normalized to the Reference cycle count. Blue is the fraction of baseline cycles remaining in Integrated; the light segment is the conservative reduction. Warm and cold decapsulation are distinct workloads and are not averaged.

Table 1. The eight retained implementation transformations and their X-Wing scope. None changes the X-Wing, ML-KEM, or X25519 specification.

#	Component	Transformation	Active paths
1	ML-KEM	MVE NTT with consumer-order structure stores	KG/E/W/C
2	ML-KEM	Eleven MVE data-conversion functions	KG/E/W/C
3	X25519	$w = 3$ signed comb for basepoint 9	KG/E/C
4	X25519	Fixed-frame field-multiplication schedule	KG/E/W/C
5	X25519	Equal-instruction field-squaring schedule	KG/E/W/C
6	Common	Alignment-aware word `memcpy`/`memset`	KG/E/W/C
7	X25519	Four-chunk 128-bit MVE xor-mask `cswap`	E/W/C
8	X-Wing	Zero-safe paired inversion of two X25519 endpoints	E only

Table 2. Direct same-ELF comparison of the Reference and Integrated implementations. Each cell is the median of 100 measurements. Ranges cover ten adjacent Reference/Integrated or Integrated/Reference pairs from five independent reflashes; the last column is the conservative headline.

Operation	Reference cycles	Integrated cycles	Reduction range	Reported
KeyGen	870,869–870,970	700,834–700,865	19.521–19.534%	**19.52%**
Encaps	1,326,924–1,326,931	1,064,853–1,064,862	19.750–19.751%	**19.75%**
Decaps (warm)	934,803–934,837	825,746–825,793	11.661–11.670%	**11.66%**
Decaps (cold)	1,802,626–1,802,655	1,523,463–1,523,502	15.484–15.488%	**15.48%**

2 Tables

Table 3. Leave-one-out contribution of the six transformations in one representative complete run. Each entry is the cycle increase when the named transformation is disabled from the six-transform implementation. These diagnostic values are not added to the final Reference–Integrated percentages.

Disabled axis	KeyGen	Encaps	Warm	Cold
Fixed-base signed comb	152,860	152,710	16	152,732
MVE NTT/current-order store	22,556	18,002	28,180	50,808
MVE data conversions	11,120	33,300	39,386	50,552
Aligned `memcpy`/`memset`	12,660	10,042	9,906	21,380
Field-squaring schedule	3,488	22,770	19,296	22,794
Field-multiplication schedule	4,182	11,950	7,632	11,898
Leave-one-out sum	206,866	248,774	104,416	310,164
Same-run disabled–enabled difference	226,096	268,196	104,717	329,445
Interaction residual	19,230	19,422	301	19,281

Table 4. End-to-end effect of the two additional X25519 transformations on top of the six-transform implementation. Values are ranges from two independent reflashes in a separate same-ELF experiment and are diagnostic rather than additive to Table 2.

Transformation/path	Baseline cycles	Candidate cycles	Reduction
Encapsulation			
MVE `cswap`	1,128,441–1,128,448	1,114,176–1,114,177	1.264–1.265%
Paired inversion	1,128,441–1,128,448	1,095,069–1,095,076	2.957%
Both	1,128,441–1,128,448	1,080,740–1,080,749	**4.226–4.228%**
Decapsulation with MVE `cswap`			
Warm	839,319–839,328	824,987–824,991	1.707–1.709%
Cold	1,553,257–1,553,270	1,538,838–1,538,852	0.928%

Table 5. Microarchitectural candidates and their end-to-end adoption decisions. A local win is retained only when it survives its specified complete-path gates.

Candidate	Micro result	Complete-path observation	Decision
MVE `cswap`	46.851% faster	Encaps 1.264–1.265%; warm 1.707–1.709%; cold 0.928% faster	Retain
Paired inversion	47.487% faster	Encaps 2.957%; 4.226–4.228% with `cswap`	Retain
Direct-64 Keccak	356.545% slower	Code expansion and spills fail the micro gate	Reject
MVE rejection/MAC	51.460% slower	Predicate, spills, and scalar compaction dominate	Reject
Dual-beat `cswap` schedule	12.883% faster	Cold improves only 0.162–0.163%, below the 0.20% gate	Reject

Table 6. Measured time–space trade-off of the separate six-transform implementation. The dispatcher-free Integrated footprint remains outside the measured scope.

Item	Measured cost or interpretation
Six transforms: nonvolatile-memory increase	+34,308 B
Six transforms: static-RAM increase	+2,040 B
Fixed-base table	33,024 B; 96.25% of the measured nonvolatile increase
Removing the fixed-base table	Approximately +152k cycles in key generation, encapsulation, and cold decapsulation
Six-transform comparison stack	Disabled and enabled configurations are equal per path; no 64-B guard corruption
Final comparison stack	Reference and Integrated are equal per path with the common comparison wrapper